జాతీయ సాంకేతిక విజ్ఞాన సంస్థ వరంగల్
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# Parking Survey Report

Display Details of Project

**Table 1: ECS Values as per IRC SP 12**

| Vehicle Type | ECS |
|---|---|
| Two Wheeler | 0.25 |
| Car/Taxi | 1 |
| Auto Rickshaw | 0.5 |
| Bicycle | 0.1 |
| Trucks/Buses | 2.5 |
| Emergency Vehicles | 2.5 |
| Rickshaw | 0.8 |

| Slot | Two Wheeler | Car/Taxi | Auto Rickshaw | Bicycle | Trucks/Buses | Emergency Vehicles | Rickshaw | ECS |
|---|---|---|---|---|---|---|---|---|
| 0.00-0.15 | 139 | 38 | 0 | 0 | 1 | 0 | 0 | 75.25 |
| 0.15-0.30 | 144 | 40 | 0 | 0 | 1 | 0 | 0 | 78.5 |
| 0.30-0.45 | 145 | 42 | 0 | 0 | 1 | 0 | 0 | 80.75 |
| 0.45-1.00 | 162 | 44 | 0 | 0 | 1 | 0 | 0 | 87 |
| 1.00-1.15 | 161 | 44 | 0 | 0 | 1 | 0 | 0 | 86.75 |
| 1.15-1.30 | 164 | 40 | 0 | 0 | 1 | 0 | 0 | 83.5 |
| Total | 915 | 248 | 0 | 0 | 6 | 0 | 0 | 491.75 |

**Table 2: Total Volume in ECS**

Sample Calculation :- 139*0.25 + 38*1 + 0*0.5 + 0*0.1 + 1*2.5 + 0*2.5 + 0* 0.8 =75.25

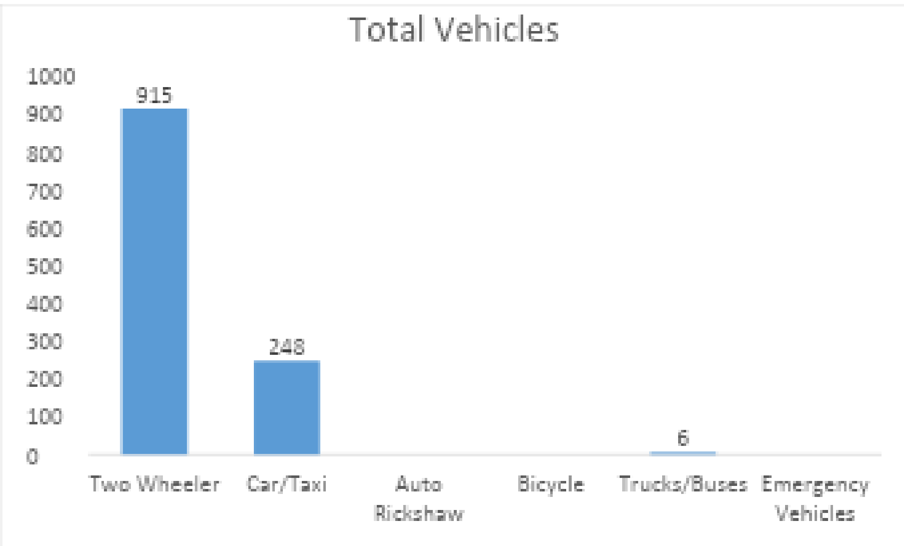


Fig-1 Total Number Vehicles in Survey

**Table 3: Area under Parking Accumulation Curve**

| X | Y | |
|---|---|---|
| 15 | 75.25 | |
| 30 | 78.5 | 1153.1 |
| 45 | 80.75 | 1194.4 |
| 60 | 87 | 1258.1 |
| 75 | 86.75 | 1303.1 |
| 90 | 83.5 | 1276.9 |
| | 491.75 | 6185.6 |

Sample Calculation :- {[30-15]/2}*[75.25+78.5] = 1153.1

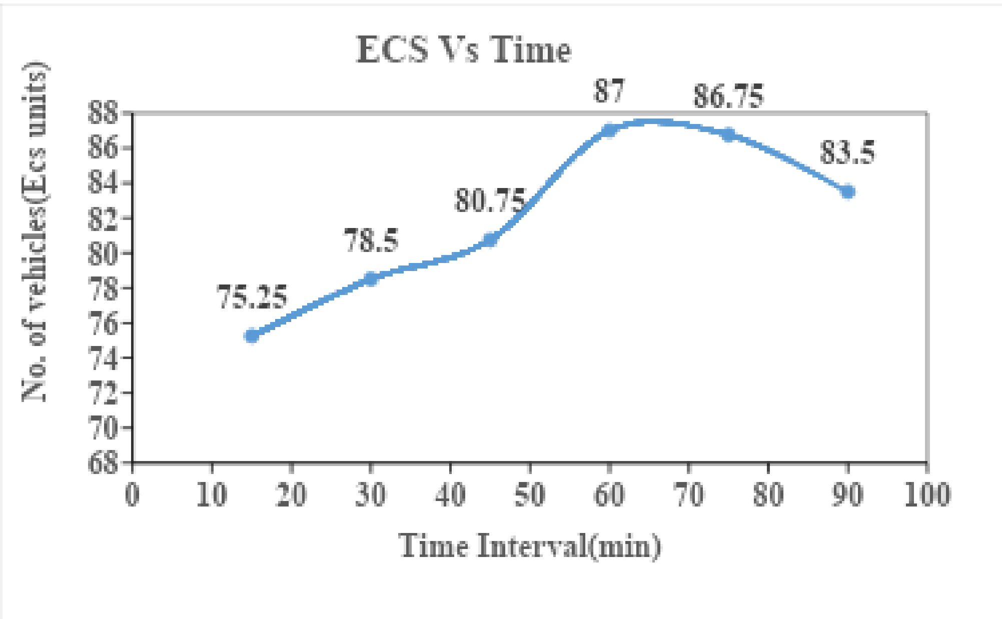


Fig-2 Parking Accumulation Curve

## Table 4: Parking Statistics

| Parking Statistics | Values | Units |
|---|---|---|
| Duration | 90 | Min |
| Parking Supply | 100 | Ecs |
| Parking Load | 6185.63 | ecs*min |
| Avg Parking Duration | 12.5788 | Min |
| | 0.20965 | hr |

Sample Calculation :- 6185.63/491.75 = 12.578

## Table 5: Parking Duration

| Vehicle | Parking Duration(hr/veh) |
|---|---|
| Two Wheeler | 0.90136612 |
| Car/Taxi | 0.887096774 |
| Auto Rickshaw | No Auto |
| Bicycle | No Bicycle |
| Trucks/Buses | 0.875 |
| Emergency Vehicles | No Emergency Vehicles |
| Rickshaw | No Rickshaw |

Sample Calculation :- {[139*1 + 144*2 +145*3 + 162*4 + 161*5 +164*6 ]*[30-15]/60}/915 = 0.901

## Table 6: Turn Over

| No of Sessions | | 6 | |
|---|---|---|---|
| Vehicle | Count | Available Space | Turn Over (veh/space/session time) |
| Two Wheeler | 915 | 150 | 1.0167 |
| Car/Taxi | 248 | 60 | 0.6889 |
| Auto Rickshaw | 0 | 5 | 0 |
| Bicycle | 0 | 5 | 0 |
| Trucks/Buses | 6 | 2 | 0.5 |
| Emergency Vehicles | 0 | 5 | 0 |
| Rickshaw | 0 | 10 | 0 |

Sample Calculation :- {(915)/(150*6)} = 1.016

Note: Available Space input should be given by user

## Table 7: Parking Supply

| | |
|---|---|
| Total duration of Survey(hr) | 1.3 |
| F | 0.9 |
| Vehicele | Parking Supply(veh) |
| Two Wheeler | 194.7044559 |
| Car/Taxi | 79.13454545 |
| Auto Rickshaw | 0 |
| Bicycle | 0 |
| Trucks/Buses | 2.674285714 |
| Emergency Vehicles | 0 |
| Rickshaw | 0 |

Sample Calculation :- {150*1.3/0.901}*0.9 = 194.704

Note: F input should be given by user

F=Insufficiency factor to account for turn over, values range from 0.85 to 0.95 and increase as average duration increases.

Table 8: Parking Index

Conclusion:

## Parking Load:

It represents the total occupancy of parking spaces throughout the study duration.

## Average Parking Duration

Average Parking Duration is the average time a vehicle remains parked.

## Turnover

Turnover indicates the number of vehicles using a single parking space during the survey period

## Parking Supply

Parking Supply represents the number of vehicles that can be accommodated during the survey period considering turnover. The available two-wheeler parking spaces can accommodate approximately **195 vehicles** during the survey period.

## Parking Index

Parking Index indicates the percentage utilization of parking spaces.